

Adroit Technologies

Protocol Driver

Name	Mitsubishi A-Series C24 Serial
DLL	MITAJC24



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1. Overview

The Mitsubishi C24 driver communicates to a AJ71C24 interface module on an A or Q series PLC, and not to the programming port.

No PLC software is required to allow communication with the AJ71C24 communication processor.

2. Wiring

2.1. RS 232 Wiring

For 25 pin male (PLC side) to 9 pin female (PC side):

AJ71C24		PC 9 pin
1	Shield	1
2		2
3		3
6		4
7		5
20		6
4 to 5 to 8		1 to 7 to 8

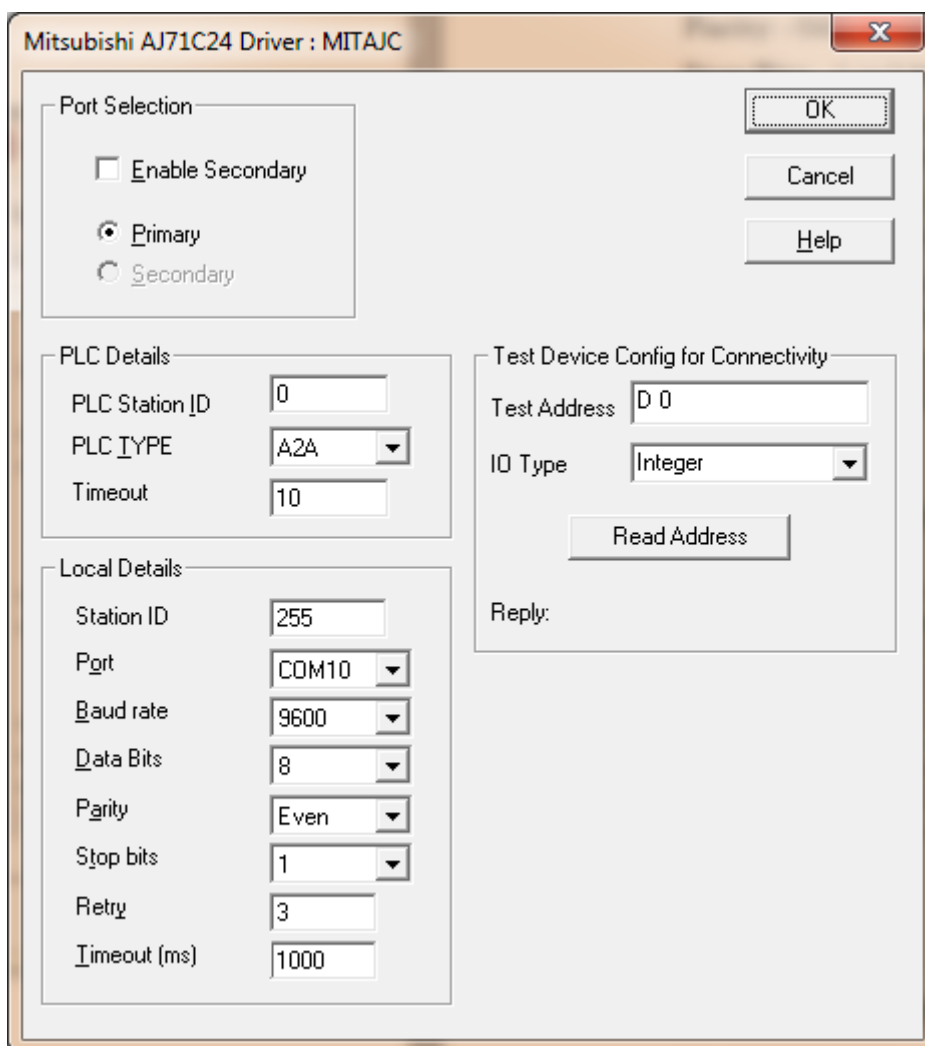
2.2. RS 422/485 Wiring

For MOXA A52 smart converter (SW 1 and 2 on):

Moxa connection	C24 pins
1	RDA
2	RDB
3	SDA
4	SDB

3. Device Configuration

3.1. Device Configuration



Mitsubishi AJ71C24 Driver : MITAJC

Port Selection

☐ Enable Secondary

☒ Primary

☐ Secondary

PLC Details

PLC Station ID: 0

PLC TYPE: A2A

Timeout: 10

Local Details

Station ID: 255

Port: COM10

Baud rate: 9600

Data Bits: 8

Parity: Even

Stop bits: 1

Retry: 3

Timeout (ms): 1000

Test Device Config for Connectivity

Test Address: D 0

IO Type: Integer

Read Address

Reply:

OK, Cancel, Help

3.1.1. Port Selection

Enable Secondary – Enables secondary configuration for dual redundant PLCs.

Primary / Secondary – Select the primary or secondary button depending on which connection you wish to configure.

3.1.2. PLC Details

PLC Station ID - The Station ID (**0 to 31 only**) for the PLC communication adapter

PLC Type - The following CPU types are supported: A1, A1N, A2, A2N, A2C, AOJ2H, A2-S1, A2N-S1, A3N, A3M, A2A, A2A-S1, A3.

Timeout - The timeout value, also referred to as the Message wait time. Please see Mitsubishi Computer Link Module documentation (SH (NA) 3511-A)

3.1.3. Local Details

Station ID – Must be 255, unless specifying another station on the MelsecNet.

Port - The serial port communication port. For example, COM1.

Baud Rate - Baud Rates between 300 and 19200 are valid.

Data Bits - 7 or 8 Data Bits.

Parity - Odd, Even or No Parity

Stop Bits - 1 or 2 Stop bits.

Retry - The number of times Adroit will attempt to recover communication in case of communication failure.

3.1.4. Test Device Config for Connectivity

This section is available to help users determine whether or not the settings they have entered are valid and Adroit can communicate with your device.

Test Address – An address in your device which will be polled when the “**Read Address**” button is clicked. The address entered must be valid.

IO Type – The address type being read. i.e. Boolean, integer, real or string.

Read Address – Click this button to poll your device with the current settings in your dialog.

Reply – When “**Read Address**” is clicked, Adroit will poll your device. The value returned from your device will be displayed here.

4. Supported Addresses

Mitsubishi A series PLC data (device) types supported in the Adroit Protocol Driver Product Version 3.51 (Pvt. Build 2)

Device		Min Range	Max Range	PLC Data Type	Boolean	Integer	Real	String
Data Reg	D	0	6143	Int 16	D 1.2	D 0	D 0	D 0 Sn
			6143	BCD 16		D 0 B	D 0 B	
			6143	Word		D 0 W	D 0 W	
			6142	BCD 32		D 0 C	D 0 C	
			6142	Long		D 0 L	D 0 L	
			6142	Real 32			D 0 R	
		9000	9255	Int 16	D 1.2	D 0	D 0	D 0 Sn
			9255	BCD 16		D 0 B	D 0 B	
			9255	Word		D 0 W	D 0 W	
			9254	BCD 32		D 0 C	D 0 C	
			9254	Long		D 0 L	D 0 L	
			9254	Real 32			D 0 R	
Link Reg	W	0	0xFFFF	Int 16	W1.2	W 0	W 0	W 0 Sn
File Reg	R	0	8191	Int 16	R 0.0	R 0	R 0	R 0 Sn
			8191	BCD 16		R 0 B	R 0 B	
			8191	Word		R 0 W	R 0 W	
			8190	BCD 32		R 0 C	R 0 C	
			8190	Long		R 0 L	R 0 L	
			8190	Real 32			R 0 R	
Ext. File Reg	E	0:0	28:8191	Int 16	E 1:0.0	E 1:0	E 1:0	E 1:0 Sn
			28:8191	BCD 16		E 1:0 B	E 1:0 B	
			28:8190	BCD 32		E 1:0 C	E 1:0 C	
			28:8190	Long		E 1:0 L	E 1:0 L	
			28:8190	Real 32			E 1:0 R	
Timer	T	0	2047	Int 16		T 0	T 0	
Counter	C	0	1023	Int 16		C 0	C 0	
Inputs	X	0	0x7FF	Bool	X 0			
			0x7F0	BCD 16		X 0 B	X 0 B	
Outputs	Y	0	0x7FF	Bool	Y 0			
Internal Rel	M/L/S	0	8191	Bool	M 0			
		9000	9255	Bool	L 0			
Link Relay	B	0	0xFFFF	Bool	B 0			

4.1. AJ71UC24 Module Settings

Set the Mode switch to mode 1 for Protocol 1 on RS-232C.

Set SW-11 to 24 as follows (settings for **19200, 8, 1, EVEN** serial communications)

SW 11	Off	RS-232C
SW 12	On	8 bits
SW 13	Off	19200
SW 14	On	19200
SW 15	On	19200
SW 16	On	Parity Enabled
SW 17	On	Parity Even
SW 18	Off	1 Stop Bit
SW 21	On	Sum Checking Set
SW 22	On	Write Enabled
SW 23	On	Computer link
SW 24	On	Ignored – Master

4.2. A1SJ71C24-R2, A1SJ71C24-PRF Module Settings

Settings for 9600,8,1,N serial communications

Mode = 1 Custom Set = On

Transmission specification setting switches

SW 3	Off	Unused
SW 4	On	Write Enabled
SW 5	On	9600
SW 6	Off	9600
SW 7	On	9600
SW 8	On	8 bits
SW 9	Off	Parity Not Set
SW 10	Off	Ignored – Odd
SW 11	Off	1 Stop Bit
SW 12	On	Sum Checking Set

4.3. A1SJ71C24-R4 Module Settings

Settings for 9600,8,1,N serial communications

Mode = A – Protocol one for RS-232C/RS-422

Transmission specification setting switches

SW 1	On	Ignored – Master
SW 2	On	Computer link
SW 3	Off	Unused
SW 4	On	Write Enabled
SW 5	On	9600
SW 6	Off	9600
SW 7	On	9600
SW 8	On	8 bits
SW 9	Off	Parity Not Set
SW 10	Off	Ignored – Odd
SW 11	Off	1 Stop Bit
SW 12	On	Sum Checking Set

OR

Settings for **9600, 7, 1, ODD** serial communications

Mode = 5

SW 8 - Off

SW 9 - On

All other SW's as above.

4.4. A2CCPUC24 and A2CCPUC24 - PRF Module Settings

Settings for 9600,8,1,N serial communications

Mode = 1 – Protocol one for RS-232C

Transmission specification setting switches

SW 11	On	9600
SW 12	Off	9600
SW 13	On	9600
SW 14	On	8 bits
SW 15	Off	Parity Not Set
SW 16	Off	Ignored – Odd
SW 17	Off	1 Stop Bit
SW 18	On	Sum Checking Set
SW 19	Off	RS-232C
SW 20	On	Write Enabled



PLEASE NOTE

Use the module document to change the serial setting to suite your needs.

Ref: Mitsubishi Electric Corporation Computer Link Module User Manual, Document Number SH (NA) 3511-A (9409) MEE

5. Protocol Definition and Considerations

5.1. Data Registers

Read Request:

ENQ	St No.	PC No.	Command	Message Wait Time	Head Device	No of Points	SumCC
	0 0	F F	W R	A	D 0010	0 2	
05	30 30	46 46	57 52	41	44 30 30 31 30	30 32	00 00

Read response:

STX	St No.	PC No.	Data	ETX	SumCC
	0 0	F F	5 6 A B 1 7 0 F		
02	30 30	0a 00	35 36 41 42 31 37 30 46	03	00 00

Acknowledge:

ACK	St No.	PC No.
	0 0	F F
06	30 30	46 46

Write Request:

ENQ	St No.	PC No.	Command	Message Wait Time	Head Device	No of Points	
	0 0	F F	W W	A	D 0010	0 1	
05	30 30	46 46	57 57	41	44 30 30 31 30	30 31	

Data	SumCC
5 6 A B	
35 36 41 42	00 00

Acknowledge:

ACK	St No.	PC No.
	0 0	F F
06	30 30	46 46

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	25-Jan-2023	K. Robinson	Changes to format, updated image resources. Added Appendix B DCOM Hardening. Removed Windows XP section	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

7.2. DLL Revision

Rev.	Date	Changes
1	13-Mar-2002	Driver only talks to a C24, wiring diagrams aligned to left.
	16-Mar-2006	Document title and page header corrected. Local station ID usage clarified,
	9-May-2006	Corrected Wiring Diagram - CCS
21	04-Jan-2021	Finalized for Adroit 10.x builds

